

Adnan Azad

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Senior Software Engineer & Technical Manager with 7+ years of experience in architecting secure, scalable enterprise systems, specializing in Java (Spring Boot), cloud-native microservices, Netflix OSS stack and API ecosystems for digital banking, fintech and trading platforms. Led complex integrations with payment gateways (JazzCash, EasyPaisa, Etisalat, etc.) contributed to transformation programs at Telenor Bank, FINCA, and Mobilink Bank. Skilled in designing real-time pipelines, reconciliation systems, and PCI-DSS/KYC-compliant solutions. Built fraud detection and reconciliation pipelines for batch and real-time systems. Expert in AWS, Docker, Kubernetes (EKS), and CI/CD optimization. Known for driving SDLC execution, performance tuning, and mentoring teams. Strong advocate of TDD, Agile practices, and engineering excellence through clean code, SOLID principles, and design patterns.

Experience

OCT 2024 – PRESENT

Manager Development | PTCL | Islamabad

- Led backend engineering using Java, Spring Boot for payment gateway integrations.
- Integrated payment gateways: JazzCash, Upaisa, EasyPaisa, Etisalat, Bank Alfalah.
- Designed and implemented batch processing applications using **Spring Batch** and **Spring Scheduler** for tasks like transaction reconciliation, settlements, and automated reporting, improving operational efficiency and accuracy.
- Developed enterprise desktop applications using **JavaServer Faces** (JSF) and **PrimeFaces** for internal business workflows, enhancing data management and user experience.
- Providing production support timely fixing production problems encountered, performing data fixes, executing batch processes & troubleshooting issues for business users and business analysts.
- Implemented **Spring AOP** for cross-cutting concerns like logging, auditing, and transaction management, ensuring cleaner and modular business logic.
- Applied algorithmic techniques for fraud detection, transaction scoring, and reconciliation logic, improving performance and decision accuracy.
- Solved complex system design challenges including distributed transaction handling, real-time batch jobs, and failover mechanisms.

MAR 2024 – OCT 2024

Senior Software Engineer | Avanza Solutions | Islamabad

- Designed and developed microservices-based systems using Java and Spring Boot, ensuring scalability and maintainability.
- Strong in event-driven architectures, Kafka messaging, and low-latency APIs, enabling reliable trading and payment operations.
- Collaborated with technical managers to thoroughly understand architectural requirements and translate them into technical implementations.
- Implemented **Spring AOP** to inject pre/post-conditions around service layers for transaction logging and security checks, enhancing observability and reducing code duplication.

- Built and documented RESTful APIs, following best practices in API design using tools like **Swagger/OpenAPI**.
- Applied microservices patterns including **API Gateway, Service Discovery, Configuration Server, and Sidecar Services**.
- Worked with technologies like **Oracle, Apache Kafka, Docker, and Kubernetes** to deliver cloud-native solutions.
- Implemented advanced security features such as **AES, RSA, TLS/MTLS, OAuth2, JWT, and session management**.
- Ensured regulatory compliance with **PCI-DSS, HIPAA, and GDPR** throughout the application lifecycle.
- Operated in **Agile/Scrum environments**, participating in sprint planning, daily stand-ups, and retrospectives.
- Deployed applications on **Tomcat and Nginx** web servers with optimized configuration for performance.
- Conducted **threat and risk assessments** to identify potential vulnerabilities and enforce secure coding practices.

JAN 2023 – MARCH 2024

Senior Software Engineer | Inbox Business Technologies | Islamabad

- Designed and developed microservices architecture using Java Spring Boot with patterns like gateway, discovery, config, and sidecar for scalability and maintainability.
- Deployed containerized applications using Docker, reducing deployment time and streamlining release cycles.
- Designed and implemented microservices architecture using Spring Boot and Netflix OSS stack (Eureka, Zuul, Ribbon, Hystrix) for service discovery, API gateway, and fault tolerance.
- Enhanced system resiliency and fault isolation using Hystrix Circuit Breaker and centralized configuration with Archaius.
- Achieved seamless service-to-service communication via Eureka registry and dynamic load balancing using Ribbon.
- Successfully designed and implemented high-performance systems capable of handling thousands of concurrent financial transactions with security and accuracy.
- Performed unit and system testing to validate functionality, reliability, and performance.
- Built and documented RESTful APIs, following best practices in API design using tools like **Swagger/OpenAPI**.
- Implemented advanced security features such as **AES, RSA, TLS/MTLS, OAuth2, JWT, and session management**.

AUG 2021 – JAN 2023

Senior Backend Software Engineer | Vialogic | Islamabad

- Designed scalable software using Java, Spring Boot, and RESTful APIs, improving performance and customer satisfaction.
- Deployed applications on AWS and implemented Docker/Kubernetes-based microservices, reducing deployment time and increasing availability.
- Optimized database queries and integrated online payment gateways, enhancing speed, reliability, and revenue.
- Led a cross-functional Scrum team (3–8 members) and collaborated with architects to align with coding standards.
- Reviewed requirements, provided technical feedback, and ensured delivery of customer-centric solutions.

FEB 2018 – JUL 2021

Full Stack Software Engineer | PEEK International | Islamabad

- Engineered end-to-end web applications using **Java** and **RESTful APIs**, combined with frontend technologies like **Angular**, **React**, and **Java Server Faces (JSF)** to deliver responsive and interactive user interfaces.
- Collaborated with architects and senior engineers to apply clean code, SOLID principles, and scalable design patterns across the stack.
- Reviewed requirements, provided technical feedback, and ensured delivery of customer-centric solutions.

Technical Skills

Languages & Frameworks: Java, Spring (Core, Boot, AOP, Security, Batch, Scheduler, Integration, Data JPA, MVC, Cloud, Actuator, Validation), Hibernate, JPA/JPA2, GraphQL, SOAP, REST, JSON, XML, Eureka (Service Discovery),

Frontend Technologies: Angular, React, Next Js, JSF (JavaServer Faces)

Cloud & DevOps: AWS (EC2, ECS, EKS, S3, SQS, SNS, RDS, Lambda, API Gateway, IAM, CloudWatch, Route 53, VPC, CloudFormation), Azure, Docker, Kubernetes, Nginx (Load Balancing), CI/CD, ELK Stack

Messaging & Streaming: Apache Kafka, ActiveMQ, MQTT, AWS SNS/SQS

Tools & Platforms: Maven, Swagger/Swagger Hub, GitHub, JIRA, JUnit, Mockito, API Gateway, Postman, IntelliJ IDEA, Eclipse, Terminus

Security & Identity: OAuth2, JWT, OpenID Connect, Keycloak, WSO2, Okta, Azure AD, TLS/MTLS, IAM Policies, RBAC, KYC, PCI-DSS, HIPAA, GDPR

Databases & Data Access: PostgreSQL, Oracle, MySQL, JDBC, Mongo

Operating Systems: Windows, Linux

Architecture Patterns: Microservices, Event-Driven Architecture, CQRS (Command Query Responsibility Segregation), Saga Pattern, API Gateway, Service Registry & Discovery, Circuit Breaker, Retry, Sidecar, Config Server, Domain-Driven Design, Multi-Tenant SaaS Architecture, Serverless Architecture, Layered Architecture, Hexagonal (Ports & Adapters), Modular Monolith, Serverless, Batch Processing, Reconciliation Systems, Low-latency transaction, Ribbon (Client-Side Load Balancing), Hystrix (Circuit Breaker), Zuul (API Gateway), Archaius (Configuration Management)

Education

FEB 2020

BS in Computer Science (Software Systems) | Federal Urdu University of Arts, Science and Technology | Islamabad

Certifications

Oracle Certified Associate Java Programmer (OCAJP) | AWS Cloud Practitioner | Master Microservices with Spring Boot, Docker & Kubernetes | Master Spring Boot Microservices with CQRS Event Sourcing | AWS Cloud Computing